

viSIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Date:		Duration:	60 minutes

Code of Conduct

I VIVETHA B 192424170 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: VIVETHA B

Annexure A – Scenario-Based Requirement Analysis

PHISHING EMAIL DETECTION AND CYBER THREAT INTELLIGENCE SYSTEM

1. Study and Analyze the Scenario

- Carefully understand the given problem statement.

The **Phishing Email Detection and Cyber Threat Intelligence System** is a cybersecurity system designed to identify and analyze phishing emails. Phishing is a cyberattack in which attackers send fake or fraudulent emails to trick users into revealing sensitive information such as usernames, passwords, bank details, or personal data. The proposed system analyzes different components of an email, including the **sender address, email content, suspicious keywords, URLs, attachments, and other email characteristics**. It identifies suspicious patterns and determines whether an email is **legitimate, suspicious, or phishing**.

The system also incorporates **Cyber Threat Intelligence (CTI)** to provide additional information about known malicious indicators such as suspicious URLs, domains, IP addresses, and other threat-related information. This helps security teams understand the nature of detected threats and take appropriate action. The main purpose of the system is to provide an **automated, intelligent, and efficient approach**

to phishing detection, reduce the risk of users falling victim to phishing attacks, and support organizations in improving their overall cybersecurity.

Scenario

When a user receives an email, the system examines the email and extracts relevant features. It then analyzes these features using phishing detection techniques. If suspicious indicators are identified, the system classifies the email as potentially malicious and generates an alert. Cyber Threat Intelligence information can then be used to provide additional details about the detected threat. Thus, the system helps organizations **detect phishing emails, analyze cyber threats, alert users, and support faster security response**.

- Identify the business domain, stakeholders, existing challenges, and system objectives.

Business Domain

The **Phishing Email Detection and Cyber Threat Intelligence System** belongs to the **Cybersecurity and Information Security domain**. The system focuses on detecting phishing emails, analyzing suspicious email content, identifying malicious links and domains, and using cyber threat intelligence to improve email security.

The system can be used by banks, educational institutions, IT companies, government organizations, healthcare organizations, and other businesses to protect users and organizational data from phishing attacks.

Stakeholders

The main stakeholders of the system are:

- **End Users** – Receive emails and use the system to identify whether an email is legitimate or phishing.
- **Security Analysts** – Analyze detected phishing emails and investigate suspicious activities.
- **System Administrator** – Manages users, system settings, detection rules, and security configurations.
- **IT/Security Team** – Monitors security alerts and takes action against detected threats.

- **Organization Management** – Reviews security reports and monitors the organization's cybersecurity status.
- **System Developers** – Develop, maintain, update, and improve the system.

Existing Challenges

The major challenges in existing phishing detection are:

- Phishing emails can appear similar to genuine emails.
- Attackers continuously create new phishing URLs and domains.
- Traditional rule-based systems may fail to detect new phishing attacks.
- Manual email analysis is time-consuming.
- Malicious links and attachments can cause security problems.
- Users may not have enough knowledge to identify phishing emails.
- Large organizations receive a high volume of emails, making manual checking difficult.
- Security teams need reliable threat intelligence to investigate suspicious emails.
- Incorrect detection can result in false positives or false negatives.
- Delayed detection can increase the damage caused by a phishing attack.

System Objectives

The main objectives of the proposed system are:

- To automatically detect phishing emails.
- To classify emails as **Legitimate, Suspicious, or Phishing**.
- To analyze sender information, email content, URLs, and attachments.
- To use Cyber Threat Intelligence to identify known malicious indicators.
- To provide alerts when a potential phishing email is detected.
- To reduce the manual effort required for email analysis.
- To improve the accuracy of phishing detection.
- To provide useful information for security analysts.
- To generate reports and statistics about detected phishing threats.
- To support faster response to cyber threats.
- To improve the overall email security of an organization.

2. Identify Functional and Non-Functional Requirements

- List all functional requirements required by the proposed system.

Functional Requirements

The **Phishing Email Detection and Cyber Threat Intelligence System** shall provide the following functional requirements:

- **User Registration and Login:** The system shall allow users to register and securely log in to the system.
- **Email Input:** The system shall allow users to upload or enter an email for phishing analysis.
- **Email Data Extraction:** The system shall extract important information such as sender address, subject, email body, URLs, and attachments from the submitted email.
- **Email Content Analysis:** The system shall analyze the content of emails to identify suspicious words, phrases, patterns, and phishing characteristics.
- **Sender Analysis:** The system shall examine the sender's email address and identify suspicious or spoofed sender information.
- **URL Analysis:** The system shall detect and analyze URLs present in emails to identify potentially malicious or phishing links.
- **Attachment Analysis:** The system shall identify suspicious attachments and flag potentially dangerous files.
- **Phishing Detection:** The system shall analyze the collected email features and determine whether the email is legitimate, suspicious, or phishing.
- **Threat Intelligence Integration:** The system shall use Cyber Threat Intelligence data to check suspicious indicators such as URLs, domains, IP addresses, and other known threat indicators.
- **Threat Indicator Identification:** The system shall identify and display threat indicators associated with a suspicious email.
- **Risk Score Generation:** The system shall generate a risk score based on the detected suspicious characteristics and threat intelligence information.
- **Result Display:** The system shall display the analysis result clearly to the user, including the classification and detected reasons.
- **Security Alerts:** The system shall generate alerts when a potentially dangerous phishing email is detected.

- **Threat Details:** The system shall provide details about detected threats to help security analysts understand the nature of the attack.
 - **Report Generation:** The system shall generate reports containing phishing detection results, risk scores, threat indicators, and analysis details.
 - **Detection History:** The system shall maintain a history of previously analyzed emails and their detection results.
 - **Dashboard:** The system shall provide a dashboard for viewing important information such as total emails analyzed, phishing emails detected, suspicious emails, legitimate emails, and threat statistics.
 - **Search and Filtering:** The system shall allow authorized users to search and filter previous email analysis results.
 - **Admin Management:** The administrator shall be able to manage users, monitor system activity, and maintain system configurations.
 - **Database Management:** The system shall securely store user information, email analysis results, threat indicators, detection history, and reports.
 - **Logout:** The system shall allow users to securely log out of their accounts.
- Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.

Non-Functional Requirements

The proposed **Phishing Email Detection and Cyber Threat Intelligence System** should satisfy the following non-functional requirements:

- **Performance:** The system should analyze an email and provide the detection result within a short time. It should process multiple email analysis requests efficiently without significant delay.
- **Security:** The system must protect user accounts, email data, threat intelligence information, and stored results from unauthorized access. Passwords and sensitive information should be securely protected.
- **Reliability:** The system should provide consistent and accurate results during email analysis. It should handle errors properly without losing important data.

- **Usability:** The system should have a simple, clear, and user-friendly interface. Users should be able to upload or submit an email and understand the detection result easily.
- **Scalability:** The system should be capable of supporting an increasing number of users, emails, and threat intelligence records without major performance degradation.
- **Maintainability:** The system should have a well-structured design that allows developers to easily update detection rules, machine learning models, threat intelligence sources, and other components.
- **Availability:** The system should be available whenever users or security teams need to analyze suspicious emails. The system should minimize downtime and recover quickly from failures.
- **Accuracy:** The system should accurately distinguish between legitimate, suspicious, and phishing emails while minimizing false positives and false negatives.
- **Compatibility:** The system should work properly with commonly used web browsers, operating systems, and standard email formats.
- **Data Integrity:** The system should ensure that stored email analysis results, threat indicators, user information, and reports are not accidentally modified or corrupted.
- **Privacy:** The system should protect confidential email content and user information and should not expose sensitive data to unauthorized users.
- **Auditability:** The system should maintain logs of important activities such as email analysis, security alerts, user actions, and administrative activities for security monitoring and investigation.

3. Identify Actors and Use Cases

- Identify all primary and secondary actors interacting with the system.

Primary Actors

The primary actors are the users who directly interact with the **Phishing Email Detection and Cyber Threat Intelligence System**.

- **End User** – Uploads or submits emails for analysis and views whether the email is legitimate, suspicious, or phishing.
- **Security Analyst** – Reviews detected phishing emails, examines threat indicators, investigates suspicious activities, and analyzes security reports.

- **System Administrator** – Manages users, system settings, access permissions, detection configurations, and system operations.

Secondary Actors

The secondary actors support the system by providing external services, data, or information.

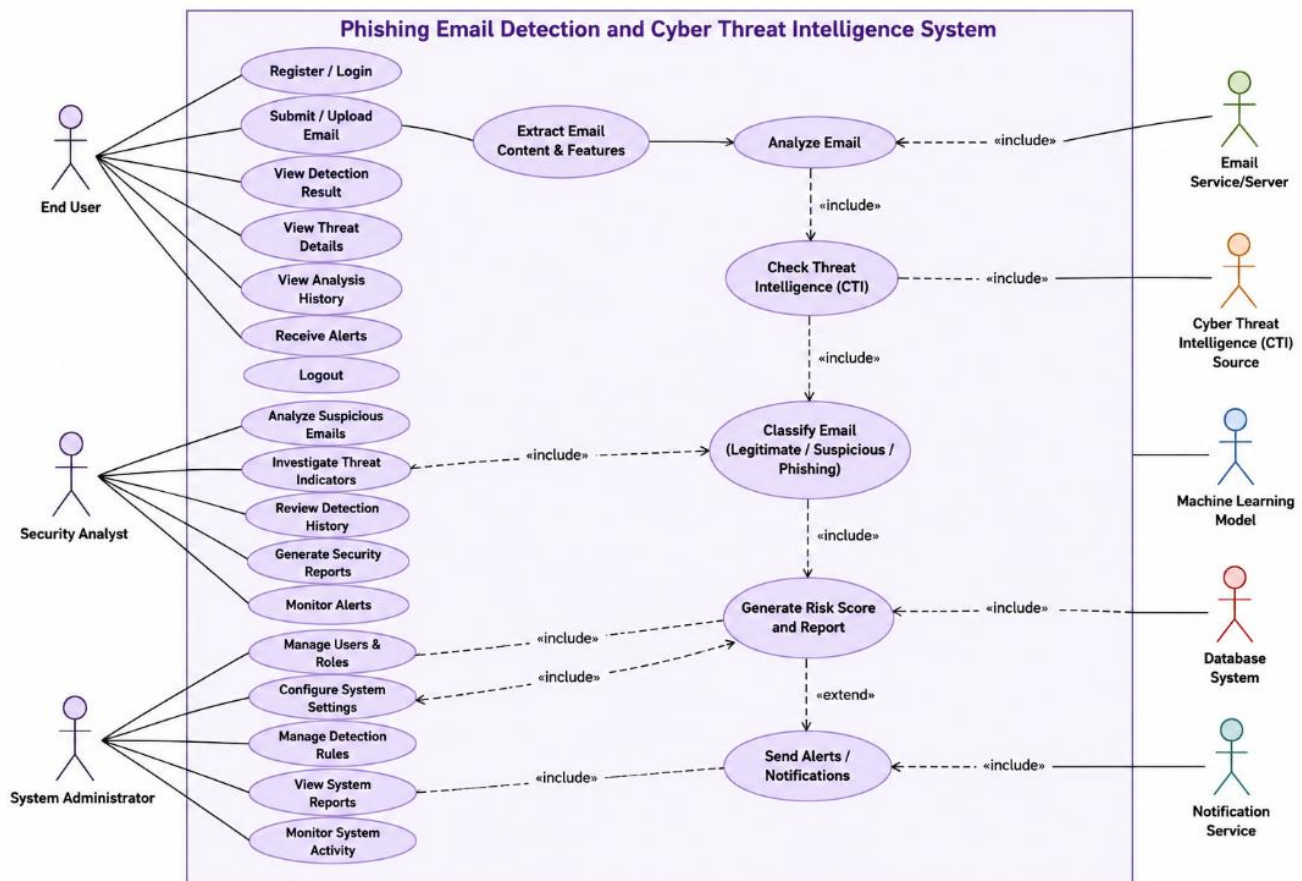
- **Cyber Threat Intelligence (CTI) Source** – Provides information about known malicious IP addresses, URLs, domains, malware, and other threat indicators.
 - **Email Service/Server** – Provides or transfers emails that need to be analyzed by the system.
 - **Machine Learning Model** – Analyzes email features and helps classify emails as legitimate, suspicious, or phishing.
 - **Database System** – Stores user information, email analysis results, threat indicators, detection history, and security reports.
 - **Notification Service** – Sends alerts or notifications when a potential phishing email is detected
- List the major use cases associated with each actor.

Major Use Cases Associated with Each Actor

Actor	Major Use Cases
End User	Register/Login, Submit Email, Upload Email File, View Detection Result, View Risk Score, View Threat Details, View Analysis History, Receive Security Alerts, Logout
Security Analyst	Login, Analyze Suspicious Emails, View Phishing Details, Investigate Threat Indicators, Check CTI Information, Review Detection History, Monitor Security Alerts, Generate Security Reports
System Administrator	Login, Manage Users, Manage Roles and Permissions, Configure System Settings, Manage Detection Rules, Monitor System Activity, Manage Threat Intelligence Sources, View System Reports
Cyber Threat Intelligence (CTI) Source	Provide Threat Intelligence Data, Update Malicious IPs, Update Malicious URLs, Update Malicious Domains, Provide Threat Indicator Information

Email Service/Server	Receive Emails, Forward Emails for Analysis, Provide Email Metadata, Provide Email Content and Attachments
Machine Learning Model	Receive Email Features, Analyze Email Features, Classify Email, Generate Phishing Probability/Risk Score, Return Detection Result
Database System	Store User Data, Store Email Analysis Results, Store Threat Indicators, Store Detection History, Retrieve Reports and Analysis Data
Notification Service	Receive Alert Request, Send Phishing Alert, Send Security Notification, Confirm Alert Delivery

- Draw a Use Case Diagram representing the interactions between actors and the system.



- Shows the interaction between **actors** and the **phishing detection system**.
- **End User** can submit emails, view results, and receive alerts.
- **Security Analyst** can investigate phishing emails and analyze threats.
- **System Administrator** manages users, settings, and detection rules.

- **CTI, ML Model, Database, Email Server, and Notification Service** support the system's operations.

4. **Prepare the Software Requirement Specification (SRS)**

Develop an SRS document containing:

- **Introduction**

The **Phishing Email Detection and Cyber Threat Intelligence System** is a cybersecurity application designed to detect phishing emails and identify potential cyber threats. The system analyzes email content, sender information, URLs, attachments, and other indicators to determine whether an email is legitimate, suspicious, or phishing. It also uses Cyber Threat Intelligence (CTI) to identify known malicious URLs, domains, IP addresses, and other threat indicators. The system helps users and security teams detect threats quickly and improve email security.

- **Problem Description**

Phishing attacks are one of the common cybersecurity threats faced by individuals and organizations. Attackers use fake emails, malicious links, fraudulent websites, and suspicious attachments to steal confidential information. Traditional detection methods may not identify newly created or sophisticated phishing attacks. Manual analysis is also time-consuming and requires security expertise. Therefore, an automated system is required to analyze emails, detect phishing attempts, provide threat information, and alert users about potential security risks.

- **Scope of the System**

The system focuses on detecting and analyzing phishing emails using email features and Cyber Threat Intelligence. Users can submit emails for analysis, and the system analyzes the sender, content, URLs, and attachments. The system classifies emails as legitimate, suspicious, or phishing and generates a risk

score. Security analysts can investigate detected threats and view reports. The system also maintains detection history and provides security alerts. The system can be extended in the future with advanced machine learning models and additional threat intelligence sources.

- **Functional Requirements**

- The system shall allow users to register and log in securely.
- The system shall allow users to upload or submit emails for analysis.
- The system shall extract sender details, subject, email body, URLs, and attachments.
- The system shall analyze email content for suspicious patterns.
- The system shall analyze URLs and domains present in emails.
- The system shall identify suspicious attachments.
- The system shall classify emails as legitimate, suspicious, or phishing.
- The system shall integrate Cyber Threat Intelligence sources.
- The system shall identify malicious URLs, domains, IP addresses, and other indicators.
- The system shall generate a risk score for analyzed emails.
- The system shall display detection results and reasons.
- The system shall generate alerts for potentially dangerous emails.
- The system shall store analysis history and threat information.
- The system shall generate security reports.
- The administrator shall be able to manage users and system configurations.

- **Non-Functional Requirements**

- **Performance:** The system should provide analysis results quickly and efficiently.
- **Security:** User accounts, email content, and threat information must be protected from unauthorized access.

- **Reliability:** The system should provide consistent results and handle errors properly.
- **Usability:** The system should have a simple and user-friendly interface.
- **Scalability:** The system should support an increasing number of users and emails.
- **Maintainability:** The system should be easy to update and maintain.
- **Availability:** The system should remain available with minimum downtime.
- **Accuracy:** The system should minimize false positives and false negatives.
- **Privacy:** Sensitive email and user information should be protected.
- **Compatibility:** The system should work with commonly used browsers and standard email formats.

- **Use Case Descriptions**

Use Case: Submit Email for Analysis

Actor: End User

Purpose: To check whether an email is phishing or legitimate.

Steps:

- User logs into the system.
- User uploads or submits an email.
- System extracts relevant email information.
- System analyzes the email content and indicators.
- System checks available threat intelligence.
- System classifies the email.
- System displays the result and risk score.

Output: Legitimate, Suspicious, or Phishing result.

Use Case: Investigate Phishing Email

Actor: Security Analyst

Purpose: To investigate a detected phishing email.

Steps:

- Analyst logs into the system.
- Analyst views detected phishing emails.
- Analyst selects a suspicious email.
- System displays email details and threat indicators.
- Analyst reviews the CTI information.
- Analyst analyzes the risk level and detection results.
- Analyst generates or reviews the security report.

Use Case: Manage System

Actor: System Administrator

Purpose: To maintain the system and manage users.

Steps:

- Administrator logs into the system.
- Administrator manages users and permissions.
- Administrator configures detection rules.
- Administrator manages system settings.
- Administrator monitors system activities.
- Administrator views system reports.

- **Data Requirements**

The system shall store and process the following data:

- **User Data:** User ID, name, email address, password, and role.
- **Email Data:** Sender address, receiver address, subject, email body, date, and email headers.
- **URL Data:** URLs, domains, URL status, and threat information.
- **Attachment Data:** File name, file type, size, and security status.
- **Detection Data:** Classification, risk score, detected features, and analysis result.
- **Threat Intelligence Data:** Malicious IP addresses, URLs, domains, and other indicators.
- **Alert Data:** Alert type, threat level, date, and notification status.
- **Report Data:** Analysis results, threat details, and security statistics.
- **History Data:** Previous email analysis and detection results.

- **External Interface Requirements**

User Interface

The system should provide a simple and user-friendly web interface containing:

- Login and registration page
- Email upload/submission page
- Email analysis page
- Detection result page
- Risk score display
- Threat details page
- Security alerts
- Analysis history

- Security dashboard
- Reports section
- Administrator dashboard

Software Interfaces

The system may interact with:

- Machine Learning models for phishing classification.
- Cyber Threat Intelligence APIs or databases.
- Database management systems for storing information.
- Email servers or email services for receiving email data.
- Notification services for sending security alerts.

Hardware Interfaces

The system should support standard computers, laptops, or servers with network connectivity. No specialized hardware is required for normal users.

Communication Interfaces

The system should use secure network communication such as **HTTPS** for transferring information between the user interface, application server, databases, and external threat intelligence services.

○ System Features

- **User Authentication:** Secure registration and login functionality.
- **Email Upload and Analysis:** Allows users to submit emails for security analysis.
- **Phishing Detection:** Detects and classifies potentially malicious emails.

- **URL and Domain Analysis:** Identifies suspicious or malicious links.
- **Threat Intelligence:** Uses CTI information to enrich threat analysis.
- **Machine Learning Classification:** Supports intelligent identification of phishing patterns.
- **Risk Score:** Provides a risk level for each analyzed email.
- **Threat Alerts:** Notifies users or security analysts about detected threats.
- **Security Dashboard:** Displays phishing statistics, detection results, and threat trends.
- **Threat Investigation:** Allows security analysts to examine suspicious emails and indicators.
- **Report Generation:** Produces reports containing analysis and security information.
- **Analysis History:** Maintains records of previously analyzed emails.
- **Administrator Management:** Allows administrators to manage users, configurations, and system activities.

5. Identify Assumptions and Constraints

- List the assumptions considered while developing the system.
- Users are assumed to have a basic knowledge of using computers and email services.
- Users are assumed to provide valid and readable email files or email content for analysis.
- The system assumes that the email data provided for analysis is available in a supported format.
- The system assumes that Cyber Threat Intelligence sources provide updated and reliable threat information.
- The system assumes that the machine learning model is trained using a suitable phishing email dataset.
- The system assumes that users have a stable internet connection while using the system.
- The system assumes that required databases and external services are available when needed.
- The system assumes that administrators regularly update the system, detection rules, and threat intelligence sources.
- The system assumes that authorized users will access confidential email and threat information responsibly.

- The system assumes that the system environment has sufficient storage and computing resources for email analysis.
 - Identify technical, operational, economic, legal, and time-related constraints affecting the project.

The following constraints may affect the development and implementation of the **Phishing Email Detection and Cyber Threat Intelligence System**:

- **Technical Constraints**

- The system requires suitable hardware, software, database, and network resources.
- Machine learning accuracy depends on the quality and size of the training dataset.
- External CTI APIs may have usage limits or require authentication.
- The system must support secure communication and protect sensitive email data.

- **Operational Constraints**

- Users need basic computer and email knowledge to operate the system.
- Regular updates are required for threat intelligence data and detection models.
- System performance may depend on internet connectivity and availability of external services.

- **Economic Constraints**

- The project may have a limited budget for cloud services, APIs, databases, and hosting.
- Paid threat intelligence services or advanced security tools may not be affordable.
- Development should preferably use open-source tools and freely available datasets where possible.

- **Legal Constraints**

- Email content may contain confidential or personal information and must be handled securely.
- The system must follow applicable data privacy and cybersecurity regulations.

- Threat intelligence data and third-party APIs must be used according to their licenses and terms of service.

- **Time Constraints**

- The system must be completed within the assigned project deadline.
- Limited development time may restrict advanced features and extensive testing.
- Testing, debugging, documentation, and deployment must be completed within the available schedule.

6. **Validate Requirement Completeness and Consistency**

- Verify whether all stakeholder requirements have been addressed.
- **End User:** Email submission, phishing detection, results, risk scores, and security alerts are addressed.
- **Security Analyst:** Threat investigation, CTI analysis, detection history, and security reports are addressed.
- **System Administrator:** User management, system configuration, access control, and monitoring requirements are addressed.
- **CTI Source:** Threat intelligence data such as malicious URLs, domains, and IP addresses is included.
- **Email Service/Server:** Email content, metadata, URLs, and attachments required for analysis are addressed.
- **Machine Learning Model:** Email classification and risk-score generation requirements are addressed.
- **Database System:** User data, email results, threat indicators, and analysis history are included.
- **Notification Service:** Phishing alerts and security notifications are addressed.

Conclusion: All major stakeholder requirements identified during the analysis have been considered and included in the SRS.

- Check for ambiguity, redundancy, inconsistency, and missing requirements.

The requirements of the Phishing Email Detection System are reviewed to identify any ambiguity, redundancy, inconsistency, or missing requirements. Ambiguous requirements should be clarified using specific and measurable terms so that they are easy to understand and implement. Redundant requirements should be identified and removed to avoid repetition. Inconsistent requirements should be checked to ensure that no two requirements conflict with each other. Missing requirements such as user authentication, data privacy, error handling, system security, prediction accuracy, response time, and model updating should also be identified and included. This validation ensures that the requirement specification is complete, clear, consistent, and suitable for system development and testing.

- Suggest improvements to enhance the quality and completeness of the requirement specification.
- Use **clear, simple, and specific language** for every requirement.
- Avoid **ambiguous words** such as “fast,” “easy,” “high,” or “secure.”
- Make each requirement **measurable and testable**.
- Remove **duplicate or redundant requirements**.
- Ensure there are **no conflicting or inconsistent requirements**.
- Include all necessary **functional and non-functional requirements**.
- Add requirements for **security and data privacy**.
- Specify **system performance**, such as detection accuracy and response time.
- Include proper **error handling and exception management**.
- Define **user roles, permissions, and authentication** clearly.
- Include requirements for **system maintenance and model updates**.
- Assign a **unique ID** to each requirement for easy tracking.
- Review requirements with **stakeholders** to ensure their needs are covered.
- Use a **Requirements Traceability Matrix (RTM)** to connect requirements with design, implementation, and testing.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Mark
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25